

# FASALGUARD

AI-Powered Agricultural Intelligence System

## CROP PREDICTION & RECOMMENDATION REPORT

### Report Details

Report ID: **FG99332189**

Generated On: **12/9/2025,**

**9:55:32 PM**

Location: **Lahore**

Forecast Period: **7 days**

Analysis Engine: **ML-Enhanced**

### AI Prediction

ML Models Used: **All 5 crops**

## EXECUTIVE SUMMARY

Based on advanced machine learning analysis of weather patterns and historical data, this report provides intelligent crop recommendations tailored to your specific location and conditions.

### Key Findings:

- Top Recommended Crop: Cotton (Score: 85%)
- Weather Outlook: Favorable conditions
- Water Availability: Irrigation required
- ML Prediction Accuracy: High confidence
- Economic Potential: Top crop shows good profit margins

## WEATHER ANALYSIS

No weather data available for analysis

## CROP RECOMMENDATIONS

## 1. Cotton - Score: 85% Ø<βÆ BEST MATCH

Predicted Yield: 4.3 tons/ha | Confidence: 65% | Water Needs: Moderate

Advantages: Temperature ideal for cotton growth, Rainfall adequate for this period

Action Plan: Ø=Û AI predicts average yield - follow standard cultivation practices, Ø<B!p Ideal temperature conditions - perfect for germination and growth

## 2. Sugarcane - Score: 65%

Predicted Yield: 52 tons/ha | Confidence: 65% | Water Needs: Moderate

Advantages: Temperature ideal for sugarcane growth, Based on regional climate patterns

Action Plan: & p AI predicts challenging conditions - implement protective measures, Consider drought-resistant varieties if rainfall is low

### 3. Maize - Score: 60%

Predicted Yield: 5.1 tons/ha | Confidence: 65% | Water Needs: Moderate

Advantages: Temperature ideal for maize growth, Ø>Ý AI predicts NORMAL yield: 5.1 tons/ha

Action Plan: Ø=Û AI predicts average yield - follow standard cultivation practices, Ø=Û Schedule irrigation every 3-4 days based on soil moisture

## MACHINE LEARNING PREDICTIONS

## CROP COMPARISON MATRIX



# **COMPREHENSIVE ACTION PLAN**

## **IMMEDIATE (1-7 days)**

- Prepare soil based on soil test
- Procure quality seeds of recommended crop
- Plan irrigation schedule
- Arrange necessary equipment

## **SHORT TERM (1-4 weeks)**

- Complete sowing/planting
- Implement irrigation plan
- Apply initial fertilizers
- Set up pest monitoring

## **MID TERM (1-3 months)**

- Regular irrigation as per schedule
- Weed management
- Pest and disease control
- Growth monitoring

## **LONG TERM (3-6 months)**

- Harvest planning
- Market linkage establishment
- Post-harvest management
- Profit analysis and planning for next season

Disclaimer: This report is generated by AI and should be used as guidance. Always consult with local agricultural experts for final decisions.

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